Automated Glaucoma Screening Using AI-Enhanced Ophthalmoscopy: Optic Disc/Cup Segmentation Across Publicly Available and Clinically Derived Fundus Images

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***Abstract*—Glaucoma is a leading cause of irreversible blindness, and vertical cup-to-disc ratio (CDR) derived from optic nerve head segmentation is a widely used structural measurement. We developed a reproducible pipeline and evaluated transfer from the ORIGA, G1020, REFUGE, and PAPILA publicly available fundus dataset collection to clinically derived images from a head-mounted ophthalmoscopy workflow at the University of Virginia (UVA). The study assembled 3,358 publicly available images using leakage-aware, dataset-specific split rules, compared three segmentation architectures, and selected a U-Net++ model with a ResNet-18 encoder. After 25 epochs of training on the publicly available datasets, the model achieved 0.842 mean foreground Dice on the 722-image fixed test split but only 0.251 patient-weighted mean foreground Dice when evaluated without adaptation on 59 clinically derived images from 20 patient/encounter groups. Clinical-only fine-tuning did not outperform its matched no-adaptation baseline. Joint training with the publicly available datasets and a clinically derived adaptation subset maintained essentially unchanged fixed-test performance at 0.844 and improved patient-weighted clinical Dice from 0.265 to 0.330 while reducing patient-weighted CDR absolute error by 0.122 on its own patient-group holdout. Image-weighted clinical Dice nevertheless decreased, and all clinical findings remain exploratory because the reference masks are approximate and the sample is small. The highest-value next steps are a larger, cleaner, segmentation-ready clinically derived dataset and repeated patient-group validation. A demonstration model trained on the publicly available datasets is released as an interactive web application.**

***Index Terms*—glaucoma, semantic segmentation, optic disc, optic cup, cup-to-disc ratio, domain shift, medical imaging, deep learning**

I. Introduction

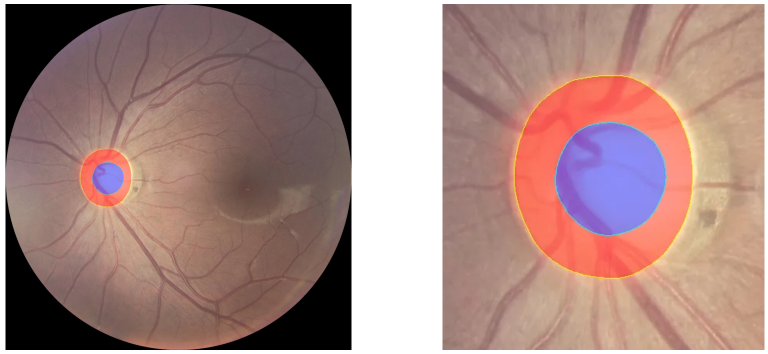
Glaucoma is among the leading causes of irreversible blindness worldwide [1], and early detection is critical because vision lost to the disease cannot be recovered. Structural assessment of the optic nerve head is one component of glaucoma screening, and vertical cup-to-disc ratio (CDR) is a widely used structural measurement associated with glaucomatous cupping. Manual delineation of the optic disc and cup is time-consuming and can vary across observers, which motivates automated segmentation from fundus imagery.

This project was proposed by the UVA Department of Ophthalmology with the long-term goal of supporting accessible structural assessment from ophthalmoscopy and reducing disparities in access to screening. The original proposal envisioned segmenting optic nerve features from ophthalmoscopy video frames and computing the Disc Damage Likelihood Scale (DDLS) [2] in real time. As the project developed and we assessed the available labeled data, we narrowed the scope to a tractable and rigorously evaluable core problem: still-image optic disc and cup segmentation and the resulting vertical CDR, which provide the anatomical foundation for later DDLS development.

We frame the work around three research questions:

- **RQ1.** How much can an optic disc/cup segmentation model trained on the publicly available fundus dataset collection be strengthened through architecture selection, online augmentation, virtual synthetic expansion, and extended training?
- **RQ2.** How well does the extended-training model derived from the publicly available fundus dataset collection transfer, without clinical adaptation, to the clinically derived dataset?
- **RQ3.** Relative to a matched no-adaptation baseline on the same patient-group holdout, how do clinical-only fine-tuning and joint training with publicly available and clinically derived data affect segmentation performance and CDR error?

Our central contribution is not a single state-of-the-art number but a reproducible account of where segmentation on the publicly available fundus dataset collection succeeds, where transfer to the clinically derived dataset fails, and which evaluated adaptation approach shows the most promising within-split evidence under realistic constraints on clinically derived data.



(a) Full fundus image with segmentation (b) Zoomed optic nerve head

Fig. 1: Illustrative optic-disc-only region (red) and optic-cup (blue) segmentation produced by the model trained on the publicly available fundus dataset collection for an image from the fixed test split, with a derived vertical CDR of 0.561. The full optic disc used for CDR is the union of the red disc-only region and blue cup. (a) The optic nerve head occupies only a small region of the full fundus frame; (b) the zoomed view shows the two predicted regions. Performance on the clinically derived dataset is substantially lower (Section V).

II. Background and Related Work

A. Optic Disc/Cup Segmentation

Fully convolutional encoder–decoder networks are widely used for biomedical segmentation. U-Net [3] introduced symmetric skip connections that recover spatial detail lost during downsampling; U-Net++ [4] redesigns those skips as nested, densely connected pathways to reduce the semantic gap between encoder and decoder features; and DeepLabV3+ [5] uses atrous spatial pyramid pooling to aggregate multi-scale context with a lightweight decoder. These architectures have been applied extensively to optic disc and cup segmentation. Fu *et al.* [6] proposed M-Net, a multi-scale U-shaped network that jointly segments disc and cup and applies a polar transformation so the two concentric structures become layered and easier to separate; the same group's Disc-aware Ensemble Network [7] combines a global-image and a disc-region branch for glaucoma screening. Across optic disc/cup segmentation studies, the cup is generally the more difficult structure to delineate because its boundary is less visually distinct than the disc boundary [6], [8]. Our work compares established architectures and then studies data-centric optimization and cross-domain transfer rather than proposing a new network.

B. CDR and Glaucoma Screening

Automated pipelines can segment the disc and cup and derive vertical CDR from their relative vertical extents. Vertical CDR is a structural measurement rather than a diagnosis; its interpretation depends on optic-disc size and other clinical findings [1], [2]. We therefore report CDR as a downstream measurement of the segmentation rather than a glaucoma classification.

C. Domain Shift and Adaptation

Models trained on publicly available benchmark datasets often degrade on clinical images acquired with different cameras, populations, and protocols [9]. Cross-domain fundus segmentation studies have reported substantial degradation when source and target images differ in acquisition device and image characteristics [10]. This domain shift is a central obstacle to applying models trained on publicly available data in clinical settings. Prior work addresses it through methods such as unsupervised or source-free domain adaptation that align source and target representations or construct target-domain pseudo-labels [9]–[11]. Those methods assume access to unlabeled target-domain images during adaptation. Our setting is more constrained—a few dozen labeled images from a single clinical site—so we evaluate whether that small labeled set is better used for clinical-only fine-tuning or incorporated into joint training with the publicly available fundus dataset collection.

D. Publicly Available Datasets and Generalizability

Consistent with the proposal's emphasis on generalizability to high-risk and underserved populations, we assembled the ORIGA, G1020, REFUGE, and PAPILA publicly available fundus datasets to broaden the range of imaging conditions represented in training. Documented demographic diversity in these datasets remains limited. Neither the publicly available fundus datasets nor the small clinically derived sample contained usable demographic labels, so demographic generalizability could not be evaluated and remains a priority for future clinical data collection.

III. Data Description

A. Publicly Available Fundus Datasets

We combined four publicly available fundus datasets obtained through Kaggle—ORIGA [12] (650 images), G1020 [13] (1,020), REFUGE [14] (1,200), and PAPILA [15] (488)—for a total of 3,358 images with optic disc and cup annotations encoded as a three-class mask. ORIGA, G1020, and PAPILA were assigned deterministic, leakage-aware group-wise 70/15/15 splits with seed 42, while REFUGE's official

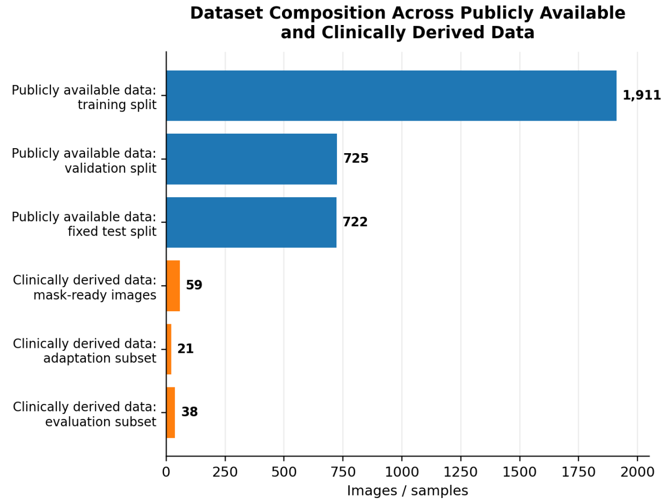


Fig. 2: Dataset composition. The publicly available fundus dataset collection contains 3,358 images divided into training, validation, and fixed test splits. The clinically derived dataset contains 59 mask-ready images; the joint-training experiment used 21 for adaptation and 38 for evaluation.

400/400/400 train/validation/test partitions were preserved. The resulting combined split contained 1,911 training, 725 validation, and 722 test images. Grouping prevents records with the same `split_group_id` from appearing in more than one split.

B. Clinically Derived Ophthalmoscopy Images

Clinical images were provided by the UVA Department of Ophthalmology as de-identified data accessed through UVA secure infrastructure under a data-use agreement and required human-subjects training. Unlike the conventional fundus-camera photographs in the publicly available dataset collection, the clinically derived images were selected still frames from a head-mounted ophthalmoscopy video workflow. These images exhibited lower resolution, video or screen-recording compression, variable focus and illumination, glare, and inconsistent framing. These acquisition differences define the cross-domain transfer problem evaluated in this study.

Disc and cup references were derived from clinically annotated PSD overlays provided by the UVA Department of Ophthalmology. The processing pipeline extracted red disc and blue cup outlines, converted them into filled approximate reference masks, and removed visible annotation marks from the model input. Of 135 processed PSD files, 59 (43.7%) yielded mask-ready image/reference-mask pairs spanning 20 patient/encounter groups. The clinically derived dataset was held entirely out of initial model development using the publicly available fundus dataset collection; for adaptation experiments it was later partitioned by patient/encounter group into adaptation and evaluation subsets. Because the derived reference masks are approximate and the sample is small, all clinical results are treated as exploratory.

IV. Methodology

A. Pipeline Overview

The project is implemented as an ordered sequence of notebooks over a shared Python package so that experiments reuse the same implementations of splitting, dataset construction, loss, metrics, and training. The pipeline proceeds in three stages: model development using the publicly available fundus dataset collection; transfer and adaptation using the clinically derived dataset; and final synthesis of externally releasable tables, figures, and claims.

B. Data Preprocessing and Manifest

All observations from the publicly available fundus dataset collection are indexed by a single manifest recording each image, its mask, its source dataset, and its split assignment. Images are loaded as RGB, resized to 256×256 (bilinear interpolation for images and nearest-neighbor interpolation for masks), and scaled to $[0, 1]$. No ImageNet mean/variance normalization is applied because the encoders are trained from scratch (`encoder_weights=None`). Masks are validated on load to contain only the labels 0, 1, and 2. Because training, evaluation, and reporting read the same manifest, split assignments remain identical across experiments.

C. Architecture Selection and Data-Centric Strategy

Under a shared five-epoch training budget on the publicly available fundus dataset collection, we compared U-Net, U-Net++, and DeepLabV3+, all implemented through the `segmentation_models_pytorch` library [16], [17]. All architecture-comparison models used 256×256 inputs, batch size 8, learning rate 10^{-4}, weight decay 10^{-4}, ResNet-18 encoders without pretrained weights, and validation mean foreground Dice for model selection. U-Net++ produced the strongest validation result and was held fixed for subsequent data-centric experiments.

We next screened online augmentation [18], including horizontal flip, small affine warps, photometric perturbation, defocus blur, low-resolution simulation, JPEG compression, and vignette illumination. Virtual synthetic expansion created one deterministic augmented view for each original training image at data-loading time, doubling the effective training rows from 1,911 to 3,822 without writing synthetic image files to disk. A later recipe combined affine, vignette-illumination, and defocus transformations with virtual synthetic expansion before the selected U-Net++ configuration was trained for 25 epochs.

The selected five-epoch U-Net++ configuration received the first evaluation on the fixed test split after validation-based model selection. The later 25-epoch retraining and joint-training experiments reused that same fixed test split for follow-up comparisons between model stages. Test images were never used for gradient updates or checkpoint selection; however, the later test results are not independent confirmatory estimates.

D. Loss and Evaluation Metrics

Training minimized a combined multiclass Dice [19] and cross-entropy loss. For a predicted binary region P and reference region G, the Dice coefficient is $\mathrm{Dice} = 2|P \cap G|/(|P| + |G|)$, ranging from 0 for no overlap to 1 for perfect overlap.

The integer masks use three mutually exclusive labels: 0=background, 1=optic-disc region excluding cup pixels, and 2=optic cup. For evaluation on the publicly available fundus dataset collection, class-specific Dice was calculated separately for the disc-only region and the cup, and mean foreground Dice was their arithmetic mean. Vertical CDR was calculated as the cup's vertical pixel extent divided by the full optic disc's vertical extent, where the full disc is the union of labels 1 and 2.

The clinically derived evaluation used full-disc Dice for all foreground pixels (labels > 0) and cup Dice for label 2. Metrics on the fixed test split from the publicly available datasets are image-weighted, whereas the primary clinical summaries are patient-weighted by first averaging within each patient/encounter group and then across groups. Consequently, the cross-domain mean-Dice comparison differs in both disc definition and aggregation and should be interpreted descriptively rather than as a strictly like-for-like benchmark.

E. Clinical Mask Derivation

Clinical reference masks were derived from clinically provided PSD files containing red and blue visual annotation overlays rather than separate pixel masks. The pipeline detected candidate red disc and blue cup components, fit filled ellipses to the selected contours, encoded the result using the shared $\{0, 1, 2\}$ convention, and inpainted the visible annotation marks from the model input. A sample was classified as "mask-ready" when it met minimum disc/cup area thresholds, at least 50% of the fitted cup overlapped the fitted disc, cup-to-disc area ratio was between 0.01 and 1.30, and derived vertical CDR was between 0.05 and 1.50. These are permissive extraction checks, not clinical adjudication.

F. Clinical Transfer and Adaptation

We first evaluated the selected five-epoch U-Net++ model and the extended-training U-Net++ model against the full 59-image clinically derived dataset without clinical adaptation (RQ2). For adaptation (RQ3), both experiments enforced patient/encounter-group separation. Clinical-only fine-tuning continued from the extended-training checkpoint using 3, 6, or 9 adaptation groups (16, 22, or 27 images) and evaluated each condition on the same 5-group, 19-image clinical holdout. Joint training added 10 clinically derived patient/encounter groups (21 images) to the publicly available training pool before augmentation and evaluated the resulting model on the remaining 10 groups (38 images). Because the clinical-only fine-tuning and joint-training experiments use different held-out clinical splits, each strategy is compared only with its corresponding model evaluated without clinical adaptation rather than ranked on a single cross-experiment leaderboard.

G. Reproducibility, Privacy, and Demonstration

All experiments use fixed random seeds and the shared package and produce aggregate artifacts suitable for external release. Private clinical images, file paths, patient identifiers, and image-level clinical metrics are excluded from every released output. The selected model trained on the publicly available fundus dataset collection is deployed as an interactive web application that segments an uploaded fundus image and reports its vertical CDR. The application is a research demonstration, uses weights trained only on the publicly available fundus dataset collection, reports a structural measurement rather than a diagnosis, and must not be used for clinical decision-making. Code and externally releasable artifacts are available at https://github.com/Rjashby1/Automated-Glaucoma-Screening-Using-AI-Enhanced-Ophthalmoscopy; the demonstration is available at https://huggingface.co/spaces/tyhob/Automated-Glaucoma-Screening-Using-AI-Enhanced-Ophthalmoscopy.

V. Results

A. Iterative Model Development on the Publicly Available Fundus Dataset Collection (RQ1)

Under the shared five-epoch architecture comparison, U-Net++ achieved the highest validation mean foreground Dice at 0.794, compared with 0.782 for U-Net and 0.759 for DeepLabV3+. Applying small-affine augmentation online increased U-Net++ validation mean foreground Dice to 0.812, a gain of 0.017. Adding one deterministic virtual synthetic view per original training image increased validation performance to 0.819, improving by 0.008 beyond online augmentation and by 0.025 over the unaugmented U-Net++ configuration. A later affine/vignette/defocus combination with virtual synthetic expansion reached 0.826 validation mean foreground Dice, a small additional short-schedule gain. These experiments establish the sequence by which architecture selection and data-centric optimization strengthened the model before extended training.

The selected five-epoch U-Net++ model achieved 0.818 mean foreground Dice in the first evaluation on the fixed test split. Extending the selected training recipe to 25 epochs increased fixed-test mean foreground Dice to 0.842. The U-Net++ model jointly trained with the publicly available dataset collection and a clinically derived adaptation subset produced 0.844 on the same fixed test split, which is effectively unchanged relative to 0.842 in the absence of repeated-seed uncertainty estimates (Fig. 3, Table I). Disc-only Dice remained higher than cup Dice at each model stage.

B. Transfer to the Clinically Derived Dataset (RQ2)

The domain gap is the headline result. The extended-training U-Net++ model achieved 0.842 mean foreground Dice on the fixed test split from the publicly available datasets but only 0.251 patient-weighted mean foreground Dice when evaluated without clinical adaptation on the full clinically derived dataset. Because the two summaries differ in disc definition and aggregation, the numerical gap is descriptive rather than

TABLE I: Performance on the same fixed test split from the publicly available fundus dataset collection.

Model and training data	Mean FG	Disc-only	Cup	CDR MAE
Selected U-Net++ (5 epochs; publicly available datasets)	0.818	0.840	0.796	0.064
Extended U-Net++ (25 epochs; publicly available datasets)	0.842	0.864	0.820	0.063
Jointly trained U-Net++ (publicly available datasets + clinical subset)	0.844	0.871	0.817	0.063

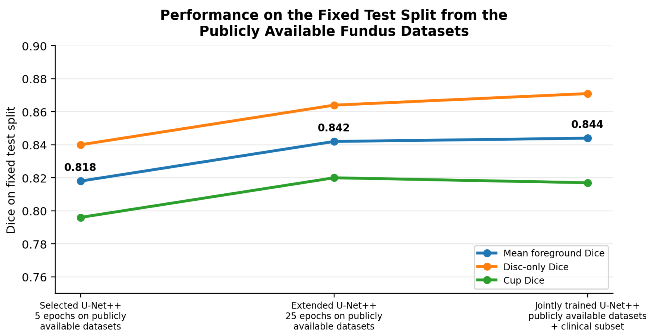


Fig. 3: Performance on the same fixed test split from the publicly available fundus dataset collection. The first test evaluation followed validation-based selection of the five-epoch model; the later stages reused the split for follow-up comparisons.

strictly like-for-like; its magnitude nevertheless indicates poor transfer to the intended acquisition domain.

The same clinically derived evaluation set provides a directly matched comparison between the selected five-epoch and extended-training models. Extending training on the publicly available datasets improved patient-weighted mean foreground Dice only slightly, from 0.242 to 0.251, while patient-weighted CDR absolute error worsened from 0.283 to 0.383 (Fig. 4a–b). Better performance on the publicly available fixed test split therefore did not produce reliable improvement on the clinically derived images.

C. Clinical Adaptation Strategies (RQ3)

Clinical-only fine-tuning did not outperform the corresponding model evaluated without clinical adaptation at any tested fraction: patient-weighted mean foreground Dice changed by -0.053, -0.044, and -0.030 for the 25%, 50%, and 75% conditions, respectively. CDR absolute error also worsened for all three conditions.

On the separate patient-group holdout reserved for the joint-training experiment, the U-Net++ model trained with both the publicly available fundus dataset collection and the clinically derived adaptation subset improved patient-weighted mean foreground Dice by 0.065, from 0.265 to 0.330, and reduced patient-weighted CDR absolute error by 0.122, from 0.385 to 0.263 (Fig. 4c, Table II). The result was not uniformly positive: image-weighted mean foreground Dice decreased from 0.382 to 0.325, although image-weighted CDR error improved. Joint training was therefore the only evaluated adaptation approach that improved its matched patient-weighted baseline, but it is not established as a general solution.

VI. Discussion

The iterative experiments show that architecture choice and data-centric optimization both mattered on the publicly available fundus dataset collection. U-Net++ outperformed the evaluated U-Net and DeepLabV3+ configurations under the shared short training budget, small-affine augmentation improved the selected architecture, virtual synthetic expansion produced a further gain, and extended training produced the clearest increase on the fixed test split.

The extended-training U-Net++ model nevertheless achieved only 0.251 patient-weighted mean foreground Dice on the clinically derived dataset. The cross-domain comparison with 0.842 on the publicly available fixed test split is descriptive because disc definitions and weighting differ, yet the matched clinical comparison reaches the same practical conclusion: extending training on the publicly available datasets changed clinical Dice only from 0.242 to 0.251 and worsened CDR error. Benchmark improvement on the publicly available datasets therefore overstated readiness for the head-mounted ophthalmoscopy imagery.

Clinical-only fine-tuning underperformed its matched no-adaptation baseline at every tested fraction. Joint training with the publicly available fundus datasets and a clinically derived adaptation subset left fixed-test performance essentially unchanged, improved patient-weighted clinical Dice, and reduced patient-weighted CDR error on its own holdout. However, image-weighted clinical Dice declined. The evidence therefore identifies joint training as the only evaluated adaptation approach that improved its matched patient-weighted baseline, not as a proven general solution.

Several limitations bound these conclusions. Clinical references were approximate ellipse-fitted masks extracted from colored PSD overlays rather than independently adjudicated pixel masks. Only 59 images from 20 patient/encounter groups were mask-ready; the two adaptation experiments used different patient-group holdouts; and disc Dice was defined differently across the publicly available and clinically derived datasets. Usable demographic labels were unavailable, so demographic generalizability remains untested.

Each model configuration was trained once under a fixed random seed, and each adaptation experiment used a single patient-group partition. Confidence intervals, repeated-seed estimates, repeated patient-group cross-validation, and statistical significance tests were not computed. Small differences between model stages—particularly 0.842 versus 0.844 on the fixed test split—should therefore be interpreted as effectively equivalent rather than evidence of statistically reliable superiority.

VII. Conclusion and Future Work

We delivered a reproducible optic disc/cup segmentation pipeline and a controlled evaluation of transfer from publicly

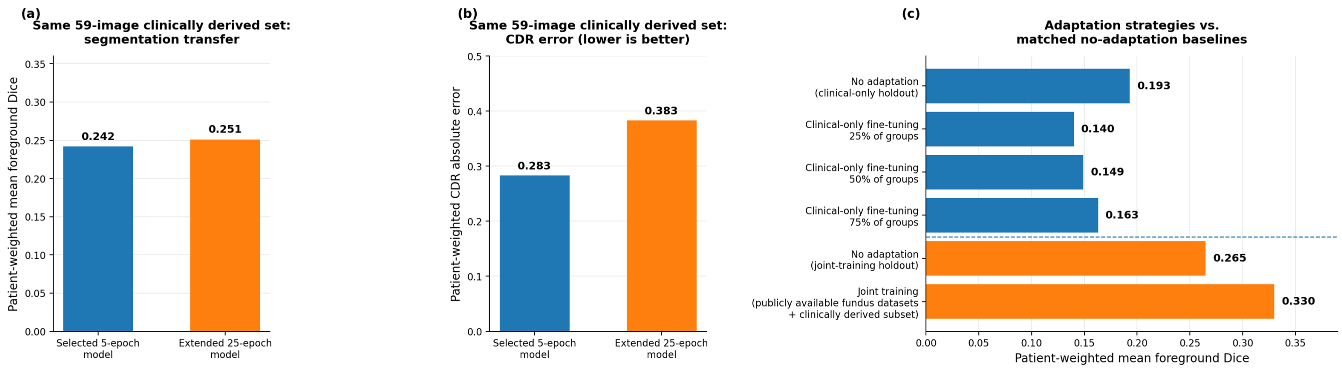


Fig. 4: Matched evaluations on the clinically derived dataset. (a) Patient-weighted mean foreground Dice and (b) patient-weighted CDR absolute error for the selected five-epoch and extended 25-epoch models on the same full 59-image clinical set. Extended training produced only a small Dice increase and worsened CDR error. (c) Patient-weighted mean foreground Dice for the adaptation strategies. Clinical-only fine-tuning and joint training used different patient-separated holdouts and are interpreted only against their matched no-adaptation baselines.

TABLE II: Patient-weighted results on the clinically derived dataset. Each strategy is compared only with its matched no-adaptation baseline because the patient-group holdouts differ. A negative change in CDR absolute error indicates improvement.

Condition	Pt.-wt. Dice	Δ Dice	Pt.-wt. CDR MAE	Δ CDR error
No adaptation on clinical-only fine-tuning holdout	0.193	0.000	0.351	0.000
Clinical-only fine-tuning using 25% of adaptation groups	0.140	-0.053	0.543	$+0.192$
Clinical-only fine-tuning using 50% of adaptation groups	0.149	-0.044	0.525	$+0.174$
Clinical-only fine-tuning using 75% of adaptation groups	0.163	-0.030	0.495	$+0.144$
No adaptation on joint-training holdout	0.265	0.000	0.385	0.000
Joint training with publicly available datasets and clinical subset	0.330	$+0.065$	0.263	-0.122

available fundus photographs to clinically derived ophthalmoscopy images. The 25-epoch U-Net++ model achieved 0.842 mean foreground Dice on the fixed test split from the publicly available datasets but only 0.251 patient-weighted mean foreground Dice without clinical adaptation. Clinical-only fine-tuning did not improve its matched baseline. Joint training maintained essentially unchanged fixed-test performance at 0.844 and improved patient-weighted clinical Dice from 0.265 to 0.330 while reducing patient-weighted CDR absolute error by 0.122 on its own holdout, although image-weighted clinical Dice declined. The system remains exploratory and is not clinically deployable.

The highest-value next step is a larger, segmentation-ready clinically derived dataset with separate, reviewable disc and cup masks. Follow-on work should use repeated patient-group cross-validation, repeated training seeds, harmonized metric definitions, pretrained encoders, and stronger domain-adaptation methods. Future work should also include automated frame-quality assessment so that poorly focused, heavily compressed, glare-obscured, or incompletely framed ophthalmoscopy images can be rejected before segmentation. Video-based evaluation and progression from CDR toward DDLS should follow only after segmentation transfer is reliable and clinically reviewed.

STUDENT CONTRIBUTIONS

Robert Ashby: Led project repository architecture and version control, data engineering, model-development. Co-developed the augmentation implementation strategy and review. Facilitated the infrastructure, experiment execution, evaluation, and pipeline reproducibility. Authored progress reporting, and sponsor-facing technical communication.

Tyler Hobbs: Developed and deployed the interactive Hugging Face demonstration. Co-developed the neural net usage strategy and review. Prepared the initial manuscript draft, and supported sponsor coordination and project review.

Emmanuel Gyamfi: Co-developed the augmentation implementation strategy and review. Supported sponsor coordination, meeting logistics, project discussions, and review of project materials.

Michael Ieraci: Co-developed the neural net usage strategy and review. Participated in team discussions and review of project materials.

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